

JENKINS

Jenkins is an open-source automation server used to automate the software development lifecycle (SDLC), including building, testing, and deploying applications. It is one of the most widely used Continuous Integration and Continuous Delivery (CI/CD) tools in DevOps.

- Jenkins is a CI/CD tool used in DevOps.
- It automates repetitive tasks such as build, test, and deployment.
- It helps detect errors early in the development process.
- It integrates with tools like Git, Docker, and AWS.
- Jenkins provides a web-based interface for managing jobs and pipelines.
- It reduces manual effort and speeds up software delivery.

Features of Jenkins

- Open-source and free to use.
- Supports thousands of plugins for tool integration.
- Pipeline as Code using Jenkinsfile.
- Real-time build monitoring and reporting.

Uses of Jenkins

- Automates software build processes.
- Supports Continuous Integration (CI) and Continuous Delivery (CD).
- Performs automated testing of applications.
- Integrates with tools like Git, Docker, AWS, and Kubernetes.

JENKINS ARCHITECTURE COMPONENTS

Jenkins Controller (Master)

- Manages Jenkins jobs and configurations.
- Schedules build tasks.
- Monitors build execution.

Jenkins Agent (Slave)

- Executes build, test, and deployment tasks.

- Can run on different machines.

Jenkins Pipeline

- Defines CI/CD workflows using a Jenkinsfile.
- Automates the complete software delivery process.

Plugins

- Extend Jenkins functionality.
- Integrate with GitHub, Docker, AWS, Maven, Kubernetes, etc.

Common Jenkins Ports

- **8080** → Default Jenkins Web UI Port
- **50000** → Agent Communication Port

Jenkins Directory Locations (Linux)

/var/lib/jenkins

HANDS-ON FOR JENKINS INSTALLATION AND PIPELINE CREATION ON UBUNTU EC2

Screenshot: Create Ubuntu EC2 Instance

- Open AWS EC2 Console.
- Launch a new Ubuntu Server instance.
- Ubuntu instance created successfully.

EC2

Instances

Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

jenkins-server

Add additional tags

▼ Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q

Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

▼ Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd6...read mo

ami-07a00cf47dbbc844c

Virtual server type (instance type)

c7i-flex.large

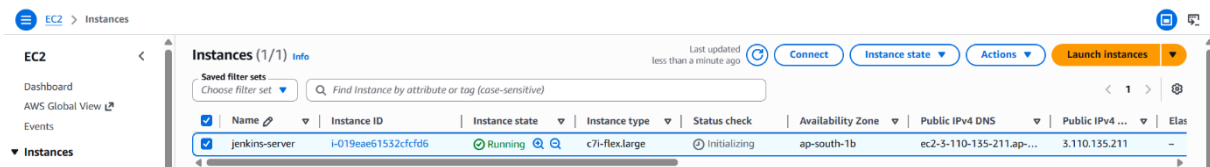
Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel



Screenshot: Switch to Root User and Update Ubuntu Server

- Connect to the Ubuntu EC2 instance.
- Execute `sudo su`.
- Switch to the root user successfully.
- Execute `apt update -y`.
- Package information updated successfully.

```
ubuntu@ip-172-31-2-102:~$
ubuntu@ip-172-31-2-102:~$ sudo su
root@ip-172-31-2-102:/home/ubuntu# apt update -y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
```

Screenshot: Install Fontconfig and Java

- Install Fontconfig using `apt install fontconfig -y`.
- Install Java using `apt install openjdk-17-jdk -y`.
- Verify Java installation using `java -version`.
- Fontconfig and Java installed successfully.

```
root@ip-172-31-2-102:/home/ubuntu# sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
39 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  fontconfig openjdk-21-jre

Installing dependencies:
  adwaita-icon-theme  gsettings-desktop-schemas  libavahi-common3  libglib2.0-0  libibus-1.0-5
  alsa-topology-conf  gtk-update-icon-cache  libcairo-gobject2  libglib2.0-dev  libice6
```

Screenshot: Install Jenkins

- Update the package information.
- Install Jenkins using `apt install jenkins -y`.
- Jenkins installed successfully.

```

root@ip-172-31-2-102:/home/ubuntu#
root@ip-172-31-2-102:/home/ubuntu# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
--2026-06-02 03:50:07-- https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:31::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring.asc      100%[=====

```

Screenshot: Copy Public IP Address

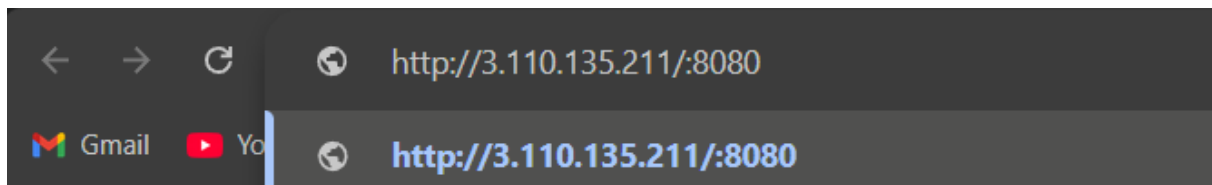
- Copy the Public IP address.
- Public IP address copied successfully.

i-019eae61532cfcfd6 (jenkins-server)

PublicIPs: **3.110.135.211** PrivateIPs: 172.31.2.102

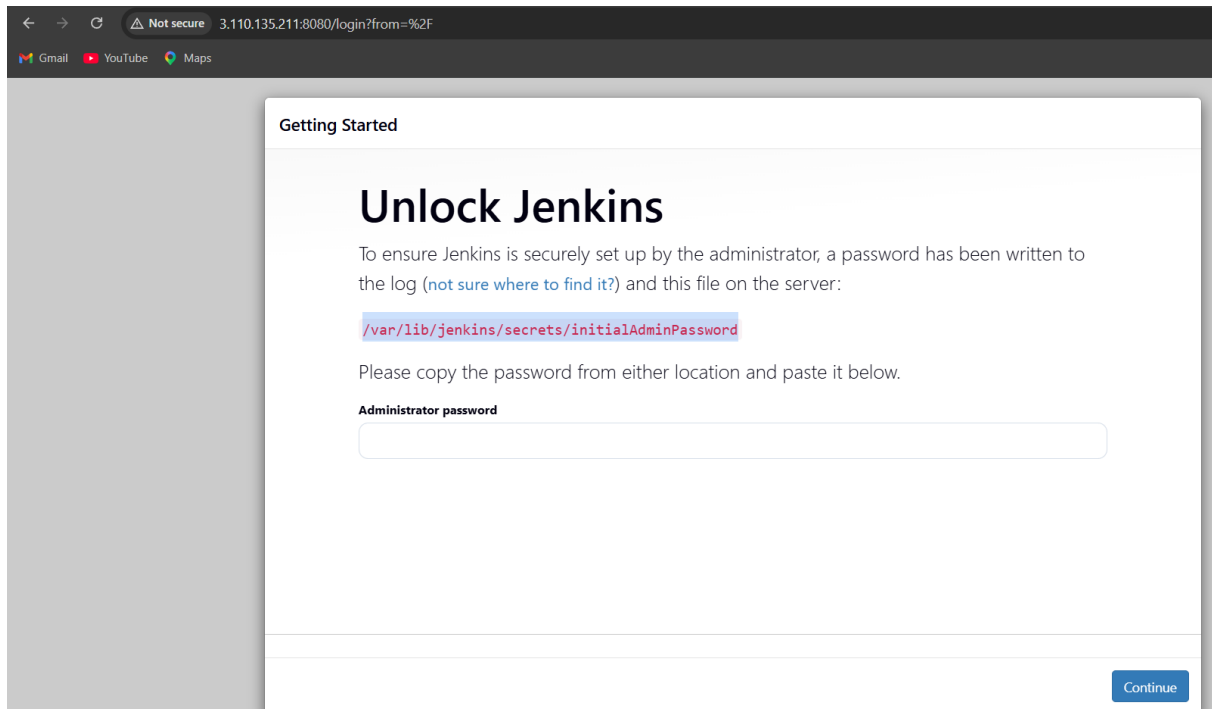
Screenshot: Access Jenkins Web UI

- Open a new browser tab.
- Enter <Public-IP>:8080.
- Jenkins unlock page displayed successfully.



Screenshot: Retrieve Initial Admin Password

- Copy the password path.
- Open the terminal and access the path.
- Copy the initial admin password successfully.



Screenshot: Retrieve Initial Admin Password

- Execute `cat /var/lib/jenkins/secrets/initialAdminPassword`.
- Copy the displayed password.
- Initial admin password copied successfully.

```
root@ip-172-31-2-102:/home/ubuntu#  
root@ip-172-31-2-102:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword  
5b34feb2c6654ec985af0a1ff949948b  
root@ip-172-31-2-102:/home/ubuntu#
```

Screenshot: Unlock Jenkins

- Paste the initial admin password.
- Click **Continue**.
- Jenkins unlocked successfully.

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Screenshot: Install Suggested Plugins

- Click **Install Suggested Plugins**.
- Jenkins installs the recommended plugins.
- Plugin installation completed successfully.

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Screenshot: Getting Started with Jenkins

- Wait for the plugin installation to complete.
- Jenkins configures the installed plugins.
- Getting Started setup completed successfully.

Getting Started

Getting Started

✓ Folders	✓ OWASP Markup Formatter	🔍 Build Timeout	🔍 Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API
🔍 Timestampers	🔍 Workspace Cleanup	🔍 Ant	🔍 Gradle	Folders
🔍 Pipeline	🔍 GitHub Branch Source	🔍 Pipeline: GitHub Groovy Libraries	🔍 Pipeline Graph View	OWASP Markup Formatter ** ASM API ** JSON Path API ** Struts ** Pipeline: Step API ** commons-text API ** Token Macro
🔍 Git	🔍 SSH Build Agents	🔍 Matrix Authorization Strategy	🔍 LDAP	
🔍 Email Extension	🔍 Mailer	🔍 Dark Theme		

** - required dependency

Screenshot: Create Jenkins Admin User

- Enter the username.
- Enter the password and confirm the password.
- Enter the full name.
- Enter the email address.
- Click **Save and Continue**.
- Admin user created successfully.

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

E-mail address

Jenkins 2.555.2

[Skip and continue as admin](#)

[Save and Continue](#)

Screenshot: Save and Finish Jenkins Setup

- Review the Jenkins configuration.
- Click **Save and Finish**.
- Jenkins setup completed successfully.

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

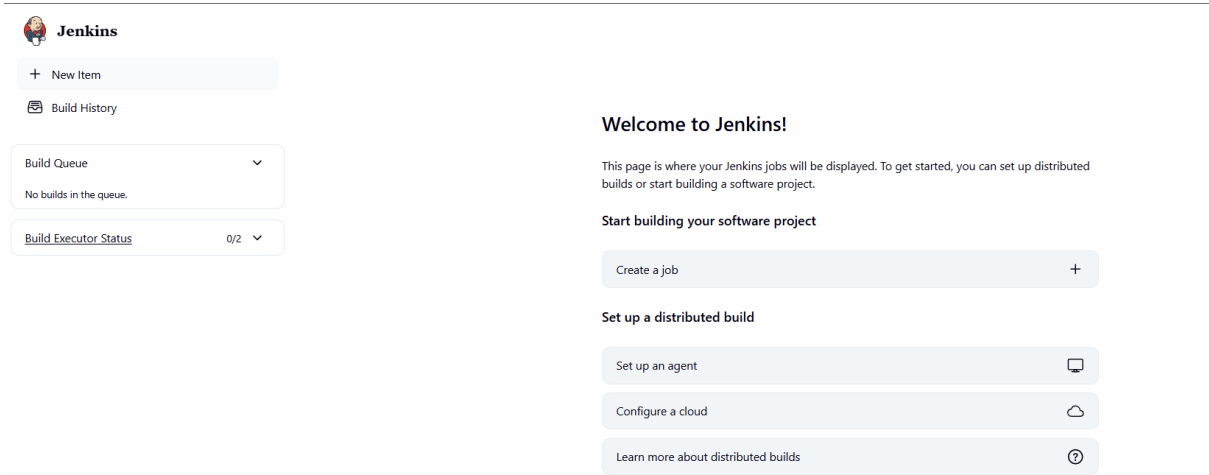
Jenkins 2.555.2

[Not now](#)

[Save and Finish](#)

Screenshot: Create New Jenkins Item

- Click **New Item** in the top-left corner.
- New Item creation page opened successfully.



Screenshot: Create Pipeline Job

- Enter a job name.
- Select **Pipeline**.
- Click **OK**.
- Pipeline job created successfully.

New Item

Enter an item name

new-work

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



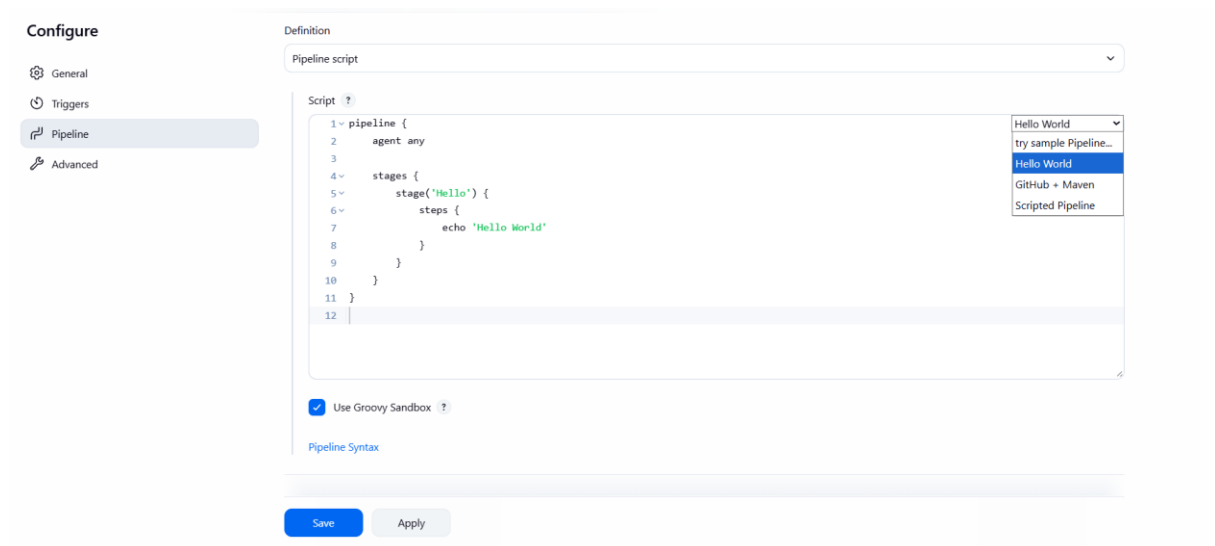
Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

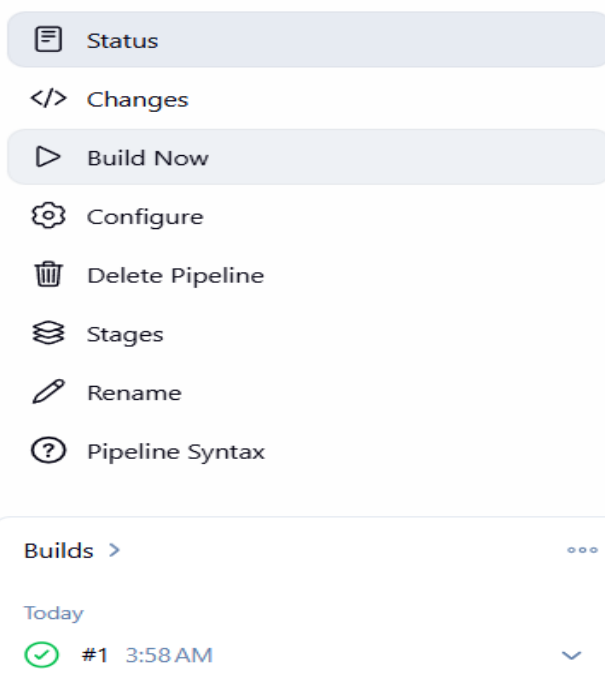
Screenshot: Configure Pipeline Script

- Scroll down to the **Pipeline** section.
- Select **Pipeline Script**.
- Enter a sample **Hello World** pipeline script.
- Click **Save**.
- Pipeline configuration saved successfully.



Screenshot: Build Pipeline Job

- Click **Build Now**.
- Pipeline build started successfully.
- Build completed successfully.



Screenshot: View Console Output

- Click the completed build.
- Select **Console Output**.

- View the pipeline execution logs.
- Hello World output displayed successfully.

Jenkins / new-work

Status

</> Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

Builds >

Filter

Today

✓ #1 3:58AM

Permalinks

- </> Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Timings
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

3 sec ago

(#1), 23 sec ago

uild (#1), 23 sec ago

uild (#1), 23 sec ago

Screenshot: Console Output

- Open the build's **Console Output**.
- Verify the pipeline execution logs.
- Output shows **Hello World** successfully.

Jenkins / new-work / #1

Status

</> Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Console

```
Started by user jai
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/new-work
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Screenshot: Verify Jenkins Job Directory

- Navigate to the Jenkins home directory.
- Open the jobs directory.
- List the available Jenkins jobs.
- new_work folder is displayed successfully.

```
root@ip-172-31-2-102:/home/ubuntu# cd /var/lib/jenkins
root@ip-172-31-2-102:/var/lib/jenkins# ls
config.xml      jenkins.install.InstallUtil.lastExecVersion  jobs      plugins      secrets
hudson.model.UpdateCenter.xml  jenkins.install.UpgradeWizard.state         logs      queue.xml    updates
hudson.plugins.git.GitTool.xml  jenkins.model.JenkinsLocationConfiguration.xml  nodeMonitors.xml  secret.key   userContent
identity.key.enc                jenkins.telemetry.Correlator.xml              org.jenkinsci.plugins.workflow.flow.FlowExecutionList.xml  secret.key.not-so-secret  users
root@ip-172-31-2-102:/var/lib/jenkins# cd jobs
root@ip-172-31-2-102:/var/lib/jenkins/jobs# ls
new-work
```